GUC Communication Theory — Assignment 1

Student Final Submission & Verification Guide

How to test in MATLAB, finalize figures, print the report, and submit

Step 1: Open and Run Mazen_Altelbany.m in MATLAB

1. Open MATLAB on your laptop or university lab PC.
2. Navigate MATLAB's Current Folder browser to this directory: e:\University\comm theory\Assignement 1.
3. Open **Mazen_Altelbany.m** and click the green **Run** button (or type Mazen_Altelbany in the Command Window).
4. The script will automatically execute, print all required tables and commentary to the Command Window, and pop up three figure windows.

Step 2: Verify the Three Output Plots

The script automatically saves the plots in high resolution to your folder:
• **task2_mae_vs_levels.png** (Figure 1: Mean Absolute Error vs L)
• **task3_variance_vs_levels.png** (Figure 2: Error Variance vs L with theoretical curve)
• **task4_sqnr_vs_levels.png** (Figure 3: SQNR vs L with theoretical curve)
These plots are already embedded in the generated Word and PDF reports. If you want MATLAB's exact native figure style, you can simply right-click the image in the Word document and select "Change Picture" → choose the MATLAB PNG.

Step 3: Review the Word Document (Assignment_1_Report_Mazen_Altelbany.docx)

1. Open **Assignment_1_Report_Mazen_Altelbany.docx** in Microsoft Word.
2. Confirm the group members listed at the top: **Mazen Altelbany (64-12371)**, **Suhad Eyhab Rasheed (64-31506)**, and **Sama Ismael Ahel (64-19880)**.
3. Check that the tables and images look clean and neatly aligned on your screen.
4. If needed, you can add any personal tweaks or print directly from Word.

Step 4: Deliverable 1 — Email the .m File

• **Recipient:** sama.ashraf@guc.edu.eg
• **Subject Line:** [COMM 601] Assignment 1 Submission - Mazen Altelbany - 64-12371
• **Attachment:** Attach only the file **Mazen_Altelbany.m**.
• **Email Body Template:**

Dear Eng. Sama,
Please find attached our MATLAB m-file for Assignment 1 (Uniform vs. K-means Quantization).
Group Members:
1. Mazen Altelbany (64-12371)
2. Suhad Eyhab Rasheed (64-31506)
3. Sama Ismael Ahel (64-19880)
Best regards,
Mazen Altelbany

Step 5: Deliverable 2 — Print the Physical Report for Office C4.329

• You can print either **Assignment_1_Report_Mazen_Altelbany.pdf** or the Word document.
• Make sure color printing is enabled so the blue, orange, and green curves on the plots are easily distinguished.
• Staple the pages at the top-left corner.
• **Submission Location:** Office **C4.329 (back)**.
• **Strict Deadline: 9 November 2026**.

Step 6: Quick Oral Evaluation Preparation

The handout explicitly states: *"Any line of code written should be fully understood by the students; otherwise, marks will be deducted during the evaluation."*

Be ready to answer these 3 common questions if the TA asks:

1. **Why is theoretical variance $\Delta^2/12$?** Because quantization error is uniformly distributed between $[-\Delta/2, \Delta/2]$. The variance of any uniform random variable of width W is $W^2/12$. Here $W = \Delta$, so $Var = \Delta^2/12$.
2. **Why does SQNR go up by ~6 dB per bit?** Because doubling L halves Δ, which cuts noise variance by 4. In decibels, $10 \cdot \log10(4) \approx 6.02$ dB.
3. **Why did K-means and Uniform give the same values?** Because the input is uniformly distributed. The optimal centroid of a uniform interval is its exact midpoint, so K-means converged to the uniform quantizer.